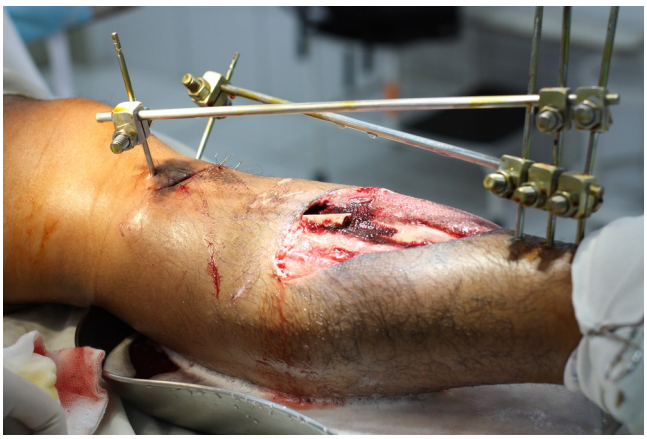
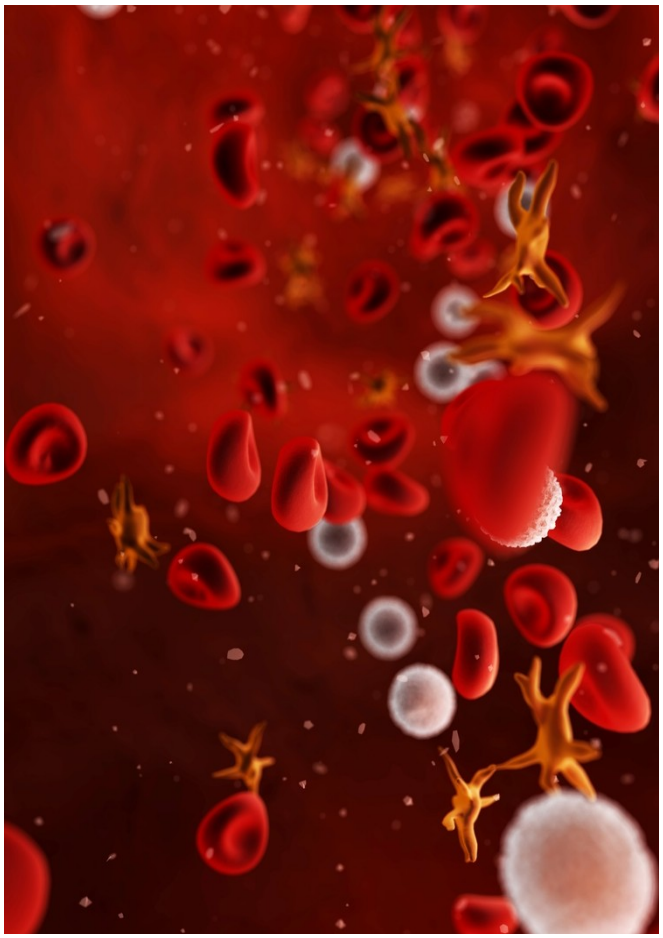
Hemostasis refers to a spectrum of events that occur as a result of damage to tissue or the blood components. The four major components to hemostasis are vasoconstriction, platelet plug formation, blood clot formation, and permanent closure of the wound with fibrous scar tissue.

Vasoconstriction

The first response to damage to a blood vessel is vasoconstriction, which limits the amount of blood lost and reduces the size of the opening that must be plugged. This response results from local myogenic spasm, release of various factors from the damaged tissues and platelets, and from nervous impulses.

In most cases, the most influential of these is myogenic spasm that results when the muscle fibers are severed during trauma. The nervous impulses usually stem from pain sensation at the site of damage. Platelets also release thromboxane A_2, which causes vasoconstriction, further reducing blood loss from small vessel damage.





Platelet Plug Formation

Platelets are anuclear cells derived from megakaryocytes in the bone marrow, are about 1-4 microns in size, and have a half-life in the bloodstream of about 1-2 weeks. The platelet cell surface is coated with glycoproteins that prevent attachment to intact endothelium, but promote adherence to damaged tissue.

When platelets come into contact with damaged endothelium, they swell and their shape becomes irregular as pseudopods extend from their surface. They begin to adhere to the exposed collagen as well as to von Willebrand's factor that is released from the damaged tissue. They release multiple factors such as thromboxane A2 and massive amounts of adenosine diphosphate (ADP), which serves to activate nearby platelets, making them adhere to each other and to the site of damage. As the platelet plug formation continues, fibrin threads begin to weave through the activated platelets, tying them together more tightly.

Platelets, along with chemical activators released from the damaged tissue and proteins within the blood stream contribute to the formation of a blood clot.

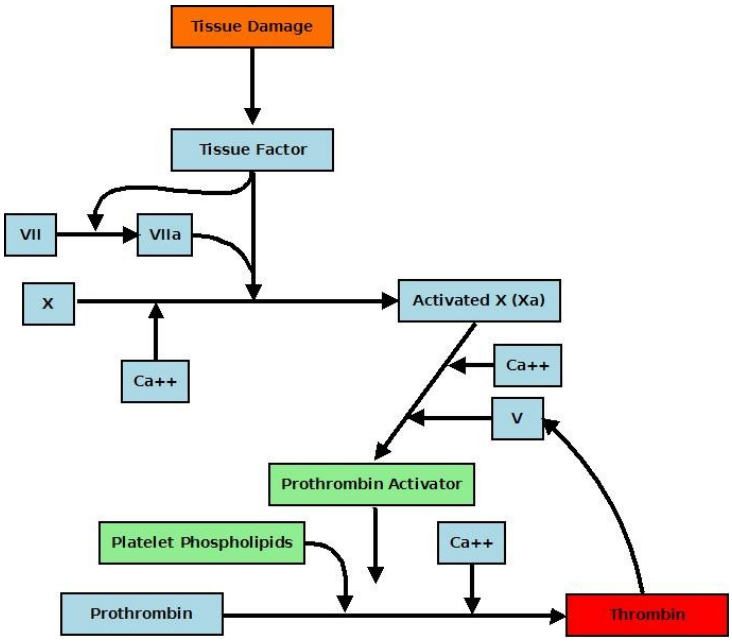
Blood Clot Formation

A blood clot forms in response to damage to a blood vessel (or damage to the blood itself) and occurs in three distinct phases: a complex series of chemical reactions known as the clotting cascade, the conversion of prothrombin into thrombin, and the conversion of fibrinogen into strands of fibrin.

There are two pathways by which clotting can be initiated: the extrinsic pathway and the intrinsic pathway. Both involve the protein factors in the table to the right. The extrinsic pathway is initiated by the release of Tissue Factor from damaged tissue. The intrinsic pathway is initiated by the activation of Factor XII and release of platelet phospholipids when the blood itself is damaged or exposed to collagen. The extrinsic pathway is the fastest, resulting in clotting in as little as 15 seconds. The intrinsic pathway is slower and usually takes between 1 and 6 minutes to produce a clot.

Both the extrinsic and intrinsic pathway result in the conversion of prothrombin to thrombin. Both pathways also require calcium at several stages of the cascade. Without calcium, neither the extrinsic or intrinsic pathway can proceed.

Factor	Name	Action
I	Fibrinogen	Forms a clot
II	Prothrombin	Activates I, V, VII, XIII, platelets, and protein C
III	Tissue Factor	Is a cofactor of VII
IV	Calcium	Promotes coagulation reactions
V	Proaccelerin	Is a cofactor of X
VI	(Unassigned)	(None)
VII	Proconvertin	Activates IX and X
VIII	Antihemophilic factor A	Is a cofactor of IX
IX	Christmas factor	Activates X
X	Stuart factor	Activates II
XI	Antihemophilic factor C	Activates IX
XII	Hageman factor	Activates XI
XII	Fibrin-stabilizing factor	Creates fibrin crosslinks



The Extrinsic Pathway

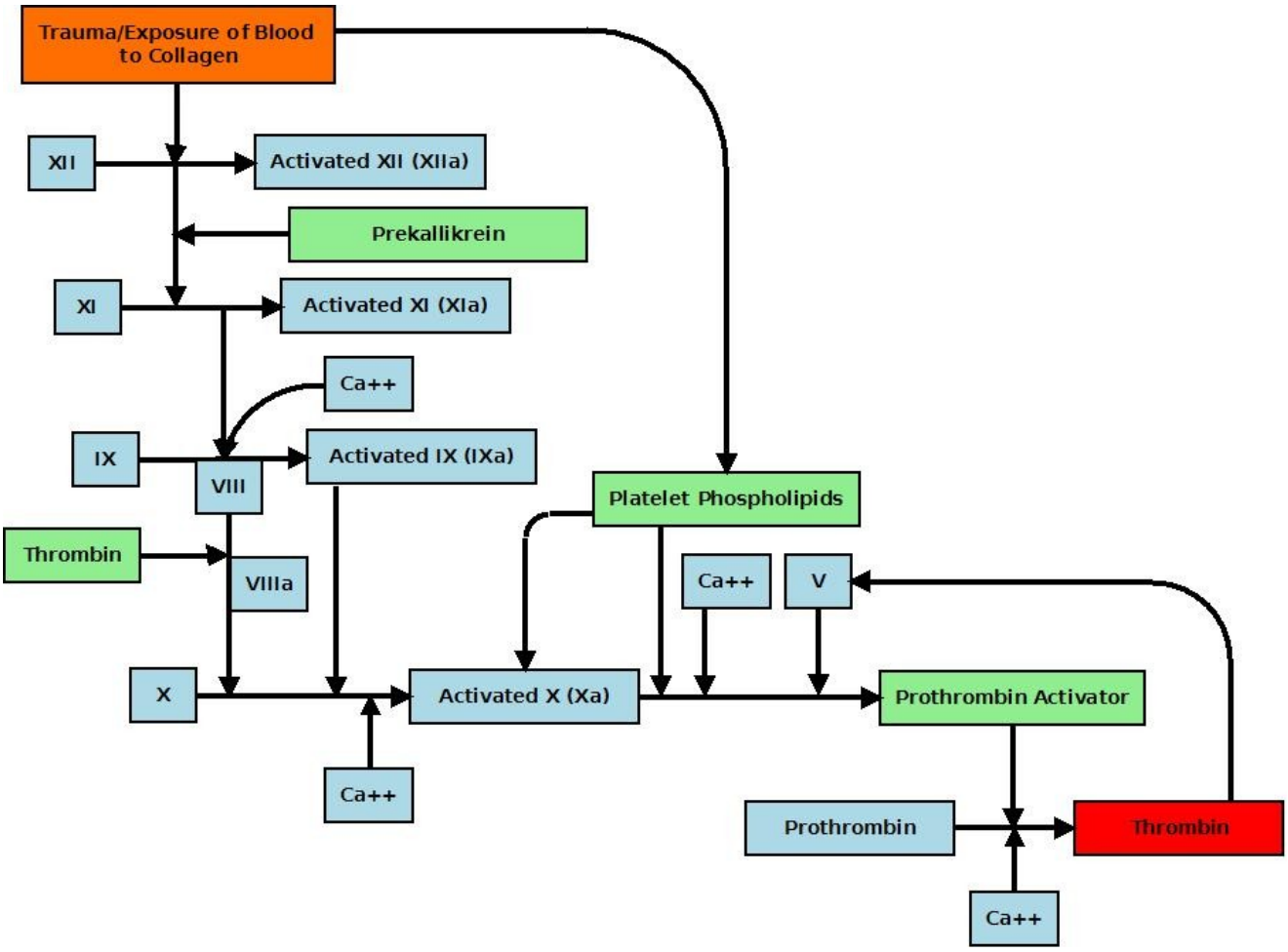
The extrinsic pathway is initiated by the release of a group of proteins known as Tissue Factor when tissues are damaged. Tissue Factor forms a complex with Factor VII, and in an enzymatic reaction requiring calcium, catalyzes the activation of Factor X.

In another reaction requiring calcium, activated Factor X combines with inactivated Factor V and phospholipid particles released from tissue and platelets to form a complex known as Prothrombin Activator. Requiring calcium a third time, Prothrombin Activator acts as a protease that splits Prothrombin into Thrombin.

Once formed, Thrombin activates Factor V. Activated Factor V increases the conversion of Prothrombin into Thrombin even more than inactive Factor V, producing a positive feedback loop by which Thrombin increases its own production. Note that the speed at which the extrinsic pathway progresses is limited only by the supplies of Factors X, VII, and V available in the bloodstream.

The Intrinsic Pathway

Clotting can be initiated by the intrinsic pathway if a blood vessel is damaged or if the blood is exposed to the collagen within a blood vessel's wall. The first step in this process is the activation of Factor XII and the release of phospholipids from the platelets. Factor XII reacts with Prekallikrein and other components like High Molecular Weight Kininogen to activate Factor XI. The phospholipids released from the platelets play a role in a later stage of the pathway.



The activated Factor XI then acts as a catalyst in the presence of calcium to activate Factor IX. Factor IX utilizes Factor VIII and platelet phospholipids to induce the activation of Factor X. It is the lack of Factor VIII at this step that produces hemophilia A.

Factor X combines with Factor V in the presence of calcium to convert to Prothrombin Activator, which is then able to split Prothrombin into Thrombin.

With the exception of the first two steps of the intrinsic pathway, calcium is required at every step of both the intrinsic and extrinsic clotting cascades. If serum ionized calcium levels fall, the clotting cascade can be impaired.

Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 451-457.